

CSE 401 – Numerical Methods

CT3: Mathematics, Derivations, Proofs & Worked Simulations

From NZR (Suchi) hand-notes

September 8, 2026

Scope. This booklet collects *only* the mathematics of the course: every derivation and proof from the notes, plus fully worked *simulations* (step-by-step numeric runs) of each algorithm so you can reproduce them under exam conditions. Theory-in-words is in the separate Q&A booklet.

Contents

1	Golden Section Search	1
1.1	Core relations	1
1.2	Proof / Derivation of the Golden Ratio	1
1.3	Worked simulation — maximize $f(x) = 2 \sin x - \frac{x^2}{10}$ on $[0, 4]$	2
2	Gradient Descent	2
2.1	Update equations	2
2.2	Deriving the SSR gradients (linear regression)	2
2.3	Worked simulation — 1-variable GD	3
2.4	Stochastic / mini-batch GD	3
3	Newton’s Method for Optimization	3
3.1	Iteration formulas	3
3.2	Derivation from the 2nd-order Taylor expansion	3
3.3	Second-order (minimum) condition	4
3.4	Worked simulation — $f(x) = x^2 - 4 \cos x$ (1-D optimization)	4
4	Foundations: Gradient & Stationary Points	4
4.1	Classifying 1-D stationary points	4
4.2	Gradient vector (2-D)	4
5	Constrained Optimization (Lagrange Multipliers)	4
5.1	Stationarity condition	4
5.2	Why the gradients are parallel (derivation)	4
5.3	Bordered Hessian	5
5.4	Worked example (full solve)	5
6	Least-Squares Linear Regression	5
6.1	Normal-equation results	5
6.2	Proof of the least-squares formulas	5
6.3	Correlation coefficient	6
6.4	Worked simulation — fit a line & find r	6

7 Multiple Linear Regression	6
7.1 Gradient of the MSE cost (derivation)	6
8 Lagrange Interpolation	7
8.1 Formula	7
8.2 Proof	7
8.3 Worked simulation — 3 points, estimate $f(2)$	7
9 Linear Interpolation	8
9.1 Derivation	8
9.2 Worked simulation	8
10 Newton's Divided-Difference Interpolation (v2)	8
10.1 Form and coefficients	8
10.2 Derivation of the first two coefficients	8
10.3 Proof of the recurrence relation	9
10.4 Worked simulation — build the table & polynomial	9
11 Spline Interpolation (v2)	9
11.1 Equation-count bookkeeping	9
11.2 Worked simulation — quadratic spline, 3 points	10
11.3 Cubic spline boundary conditions	10

1 Golden Section Search

1.1 Core relations

$$\begin{aligned}
 R &= \frac{\sqrt{5}-1}{2} \approx 0.61803, & d &= R(x_u - x_L), \\
 x_1 &= x_u - d, & x_2 &= x_L + d, \\
 x_{\text{opt}} &= \frac{x_u + x_L}{2}, & \text{stop when } \left| 2 \frac{x_u - x_L}{x_u + x_L} \right| &< \varepsilon.
 \end{aligned}$$

1.2 Proof / Derivation of the Golden Ratio

Interval lengths in one iteration:

$$\ell_0 = x_u - x_L, \quad \ell_1 = x_2 - x_L, \quad \ell_2 = x_u - x_2.$$

So that next-iteration points reuse a previous point, impose:

$$(i) \ell_0 = \ell_1 + \ell_2, \quad (ii) \frac{\ell_1}{\ell_0} = \frac{\ell_2}{\ell_1}.$$

Substitute (i) into (ii):

$$\frac{\ell_1}{\ell_1 + \ell_2} = \frac{\ell_2}{\ell_1} \Rightarrow \frac{\ell_1 + \ell_2}{\ell_1} = \frac{\ell_1}{\ell_2} \Rightarrow 1 + \frac{\ell_2}{\ell_1} = \frac{\ell_1}{\ell_2}.$$

Put $R = \ell_2/\ell_1$: then $1 + R = \frac{1}{R}$, i.e.

$$R^2 + R - 1 = 0 \Rightarrow R = \frac{-1 \pm \sqrt{1^2 - 4(1)(-1)}}{2(1)} = \frac{-1 \pm \sqrt{5}}{2}.$$

Positive root: $R = \frac{\sqrt{5}-1}{2} \approx 0.618$. ■

1.3 Worked simulation — maximize $f(x) = 2 \sin x - \frac{x^2}{10}$ on $[0, 4]$

$R = 0.61803$. Set $x_L = 0$, $x_u = 4$.

Iteration 1. $d = R(4 - 0) = 2.47214$.

$$x_1 = x_u - d = 1.52786, \quad x_2 = x_L + d = 2.47214.$$

$$f(x_1) = 2 \sin(1.52786) - \frac{1.52786^2}{10} = 1.99924 - 0.23344 = 1.76580,$$

$$f(x_2) = 2 \sin(2.47214) - \frac{2.47214^2}{10} = 1.25701 - 0.61115 = 0.64586.$$

Since $f(x_1) > f(x_2)$, the maximum is to the **left**: set $x_u = x_2 = 2.47214$. New $x_2 \leftarrow x_1 = 1.52786$ (with $f_2 = 1.76580$); only x_1 is recomputed.

Iteration 2. $d = R(2.47214 - 0) = 1.52786$.

$$x_1 = x_u - d = 0.94427, \quad f(x_1) = 2 \sin(0.94427) - \frac{0.94427^2}{10} = 1.61804 - 0.08916 = 1.52888.$$

Now $f(x_1) = 1.52888 < f_2 = 1.76580 \Rightarrow$ maximum to the **right**: set $x_L = x_1 = 0.94427$; new $x_1 \leftarrow$ old $x_2 = 1.52786$, x_2 recomputed next round.

Convergence check after Iter. 2: $\left| 2 \frac{x_u - x_L}{x_u + x_L} \right| = 2 \frac{2.47214 - 0.94427}{2.47214 + 0.94427} = 0.8947$ (still $> \varepsilon$; continue). The bracket is closing on the true max near $x^* \approx 1.4276$, $f(x^*) \approx 1.7757$.

Takeaway: after the first iteration each step needs only *one* new function evaluation, because the reassigned point already carries its stored f value.

2 Gradient Descent

2.1 Update equations

Single variable (tuning intercept b of SSR):

$$\text{step} = \frac{d(\text{SSR})}{db} \times \eta, \quad b_{\text{new}} = b_{\text{old}} - \text{step}.$$

General parameter w_j :

$$w_j \leftarrow w_j - \eta \frac{\partial J}{\partial w_j}.$$

2.2 Deriving the SSR gradients (linear regression)

Residual $r_i = y_i - (b + mx_i)$, and $\text{SSR} = \sum_{i=1}^n (y_i - b - mx_i)^2$.

$$\frac{\partial \text{SSR}}{\partial b} = \sum_{i=1}^n 2(y_i - b - mx_i)(-1) = -2 \sum_{i=1}^n (y_i - b - mx_i),$$

$$\frac{\partial \text{SSR}}{\partial m} = \sum_{i=1}^n 2(y_i - b - mx_i)(-x_i) = -2 \sum_{i=1}^n x_i(y_i - b - mx_i).$$

These two derivatives are exactly what GD plugs the current (b, m) into to get each step. ■

2.3 Worked simulation — 1-variable GD

Data: $(x, y) = (1, 2), (2, 3), (3, 5)$. Fix slope $m = 1$; tune intercept b . Loss $\text{SSR}(b) = \sum (y_i - (b + x_i))^2$. Learning rate $\eta = 0.1$. Gradient: $\frac{d\text{SSR}}{db} = -2 \sum (y_i - b - x_i)$.

Start $b_0 = 0$. Residuals: $2 - 1 = 1, 3 - 2 = 1, 5 - 3 = 2 \Rightarrow \sum = 4$. grad $= -2(4) = -8$; step $= \eta \cdot (-8) = -0.8$; $b_1 = 0 - (-0.8) = 0.8$.

$b_1 = 0.8$: residuals $1.2, 1.2, 2.2 \Rightarrow \sum = 4.6 - 3(0.8)$? recompute directly: $2 - (0.8 + 1) = 0.2, 3 - (0.8 + 2) = 0.2, 5 - (0.8 + 3) = 1.2 \Rightarrow \sum = 1.6$. grad $= -2(1.6) = -3.2$; step $= -0.32$; $b_2 = 0.8 + 0.32 = 1.12$.

$b_2 = 1.12$: residuals $-0.12, -0.12, 0.88 \Rightarrow \sum = 0.64$; grad $= -1.28$; step $= -0.128$; $b_3 = 1.248$.

The steps $(0.8 \rightarrow 0.32 \rightarrow 0.128 \rightarrow \dots)$ shrink each round — GD slows as it nears the optimum. Analytic optimum: set gradient $= 0 \Rightarrow \sum (y_i - b - x_i) = 0 \Rightarrow b^* = \frac{1}{3} \sum (y_i - x_i) = \frac{1}{3}(1 + 1 + 2) = 1.3333$. The iterates march toward 1.3333.

2.4 Stochastic / mini-batch GD

Same update, but $\partial J / \partial w_j$ is estimated from one sample (strict SGD) or a small **mini-batch** each step, instead of the whole dataset — cheaper per step and robust to redundancy in large data.

3 Newton's Method for Optimization

3.1 Iteration formulas

Root finding: $x_{t+1} = x_t - \frac{f(x_t)}{f'(x_t)}$. Optimization: $x_{t+1} = x_t - \frac{f'(x_t)}{f''(x_t)}$.

Vector form: $x_{t+1} = x_t - H^{-1}g$, with $g = \nabla f(x_t)$, $H = \nabla^2 f(x_t)$; optionally $x_{t+1} = x_t - \alpha_t H^{-1}g$.

3.2 Derivation from the 2nd-order Taylor expansion

For smooth f , near a :

$$f(x) \approx f(a) + g^\top (x - a) + \frac{1}{2}(x - a)^\top H(x - a), \quad g = \nabla f(a), \quad H = \nabla^2 f(a).$$

Expand the quadratic term, using $H^\top = H$ and $x^\top a = a^\top x$:

$$(x - a)^\top H(x - a) = x^\top Hx - x^\top Ha - a^\top Hx + a^\top Ha = x^\top Hx - 2a^\top Hx + a^\top Ha.$$

Standard quadratic form:

$$q(x) = \frac{1}{2}x^\top Hx + b^\top x + c, \quad b = g - Ha, \quad c = f(a) - g^\top a + \frac{1}{2}a^\top Ha.$$

Minimize: $\nabla q = Hx + b = 0$, so

$$x = -H^{-1}b = -H^{-1}(g - Ha) = a - H^{-1}g.$$

Hence the Newton step $x_{t+1} = x_t - H^{-1}g$. ■

3.3 Second-order (minimum) condition

$\nabla^2 q = H$. The stationary point is a minimum iff H is **positive semi-definite** (all eigenvalues real, ≥ 0); if all eigenvalues > 0 it is **positive definite**. Convex $f \Rightarrow H$ PSD.

3.4 Worked simulation — $f(x) = x^2 - 4 \cos x$ (1-D optimization)

$f'(x) = 2x + 4 \sin x$, $f''(x) = 2 + 4 \cos x$. Start $x_0 = 1$ (radians throughout).
 $x_0 = 1$: $f' = 2 + 4 \sin 1 = 2 + 3.3659 = 5.3659$; $f'' = 2 + 4 \cos 1 = 2 + 2.1612 = 4.1612$.
 $x_1 = 1 - \frac{5.3659}{4.1612} = 1 - 1.2895 = -0.2895$.
 $x_1 = -0.2895$: $\sin(-0.2895) = -0.2855$, $\cos(-0.2895) = 0.9584$; $f' = 2(-0.2895) + 4(-0.2855) = -0.5790 - 1.1420 = -1.7210$; $f'' = 2 + 4(0.9584) = 5.8336$. $x_2 = -0.2895 - \frac{-1.7210}{5.8336} = -0.2895 + 0.2950 = 0.00554$.
 $x_2 = 0.00554$: $f' = 2(0.00554) + 4 \sin(0.00554) = 0.01108 + 0.02216 = 0.03324$; $f'' = 2 + 4 \cos(0.00554) = 5.99994$. $x_3 = 0.00554 - \frac{0.03324}{5.99994} = 0.00554 - 0.00554 = 0.00000$.

The true minimum is $x^* = 0$. Notice the number of correct digits roughly *doubles* each step (x_1 : ~ 0 dp $\rightarrow x_2$: 2 dp $\rightarrow x_3$: 5 dp) — quadratic convergence. Here $f'' > 0$ throughout, so we are indeed converging to a *minimum*.

4 Foundations: Gradient & Stationary Points

4.1 Classifying 1-D stationary points

$$\begin{array}{llll} J = x^2 : & J' = 2x = 0 \Rightarrow x = 0, & J'' = 2 > 0 & \text{minimum;} \\ J = -x^2 : & J' = -2x = 0 \Rightarrow x = 0, & J'' = -2 < 0 & \text{maximum;} \\ J = x^3 : & J' = 3x^2 = 0 \Rightarrow x = 0, & J'' = 6x \text{ (sign varies)} & \text{saddle.} \end{array}$$

4.2 Gradient vector (2-D)

$$J(x) = x_1^2 + x_2^2, \quad \nabla J = \begin{bmatrix} 2x_1 \\ 2x_2 \end{bmatrix} = \mathbf{0} \Rightarrow (x_1, x_2) = (0, 0) \text{ (minimum).}$$

5 Constrained Optimization (Lagrange Multipliers)

5.1 Stationarity condition

$$\nabla(J(x) + \lambda C(x)) = 0, \quad \text{subject to } C(x) = 0.$$

5.2 Why the gradients are parallel (derivation)

At a constrained optimum the objective cannot decrease along the constraint, so its gradient has no component *along* the constraint curve — i.e. ∇J is parallel to ∇C :

$$\nabla J(x) = k \nabla C(x) \Rightarrow \nabla J(x) - k \nabla C(x) = 0 \Rightarrow \nabla(J(x) + \lambda C(x)) = 0, \quad \lambda = -k.$$

This is exactly stationarity of the Lagrangian $\pi = J(x) + \lambda C(x)$. ■

5.3 Bordered Hessian

$$H(\pi) = \begin{bmatrix} 0 & \frac{\partial C}{\partial x_1} & \cdots & \frac{\partial C}{\partial x_n} \\ \frac{\partial C}{\partial x_1} & \frac{\partial^2 \pi}{\partial x_1^2} & \cdots & \frac{\partial^2 \pi}{\partial x_1 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial C}{\partial x_n} & \frac{\partial^2 \pi}{\partial x_1 \partial x_n} & \cdots & \frac{\partial^2 \pi}{\partial x_n^2} \end{bmatrix}.$$

The (1, 1) entry is 0 because $\frac{\partial}{\partial \lambda} \left(\frac{\partial \pi}{\partial \lambda} \right) = \frac{\partial}{\partial \lambda} C(x) = 0$; the border is the constraint gradient. Used to classify the point as min / max / saddle.

5.4 Worked example (full solve)

Minimize $J = x_1^2 + x_2^2$ subject to $C = x_1 + x_2 + 3 = 0$. (If given as $C(x) = g$, first rewrite $C(x) - g = 0$.)

$$\pi = x_1^2 + x_2^2 + \lambda(x_1 + x_2 + 3).$$

Stationarity:

$$\frac{\partial \pi}{\partial x_1} = 2x_1 + \lambda = 0, \quad \frac{\partial \pi}{\partial x_2} = 2x_2 + \lambda = 0, \quad \frac{\partial \pi}{\partial \lambda} = x_1 + x_2 + 3 = 0.$$

From the first two: $x_1 = x_2 = -\lambda/2$. Substitute into the third: $-\lambda/2 - \lambda/2 + 3 = 0 \Rightarrow -\lambda + 3 = 0 \Rightarrow \lambda = 3$. Hence

$$\boxed{x_1 = -1.5, \quad x_2 = -1.5, \quad \lambda = 3.}$$

Gradient check: $\nabla J = (2x_1, 2x_2) = (-3, -3)$ and $\nabla C = (1, 1)$; indeed $\nabla J = -3 \nabla C$, parallel. ■

6 Least-Squares Linear Regression

6.1 Normal-equation results

$$\text{SSE} = \sum_{i=1}^n (y_i - mx_i - b)^2, \quad b = \frac{\sum y - m \sum x}{n}, \quad m = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}.$$

6.2 Proof of the least-squares formulas

Intercept. $\frac{\partial \text{SSE}}{\partial b} = \sum 2(y_i - mx_i - b)(-1) = 0 \Rightarrow \sum (y_i - mx_i - b) = 0$, so

$$\sum y - m \sum x - nb = 0 \Rightarrow b = \frac{\sum y - m \sum x}{n}. \quad (\text{A})$$

Slope. $\frac{\partial \text{SSE}}{\partial m} = \sum 2(y_i - mx_i - b)(-x_i) = 0 \Rightarrow \sum x_i y_i - m \sum x_i^2 - b \sum x_i = 0$. Substitute b from (A):

$$\sum xy - m \sum x^2 - \frac{\sum y - m \sum x}{n} \sum x = 0.$$

Multiply by n :

$$\begin{aligned} n \sum xy - mn \sum x^2 - (\sum y - m \sum x) \sum x &= 0, \\ n \sum xy - \sum x \sum y - m(n \sum x^2 - (\sum x)^2) &= 0, \\ \Rightarrow m &= \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}. \blacksquare \end{aligned}$$

6.3 Correlation coefficient

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}, \quad r^2 = \text{fraction of variance explained.}$$

6.4 Worked simulation — fit a line & find r

Data: $(1, 2), (2, 4), (3, 5), (4, 4), (5, 5)$, so $n = 5$. Build the sum table:

						Σ
x	1	2	3	4	5	15
y	2	4	5	4	5	20
x^2	1	4	9	16	25	55
y^2	4	16	25	16	25	86
xy	2	8	15	16	25	66

$$m = \frac{5(66) - 15 \cdot 20}{5(55) - 15^2} = \frac{330 - 300}{275 - 225} = \frac{30}{50} = 0.6.$$

$$b = \frac{20 - 0.6 \cdot 15}{5} = \frac{20 - 9}{5} = \frac{11}{5} = 2.2.$$

So $\hat{y} = 0.6x + 2.2$.

$$r = \frac{5(66) - 15 \cdot 20}{\sqrt{[5(55) - 15^2][5(86) - 20^2]}} = \frac{30}{\sqrt{50 \cdot 30}} = \frac{30}{\sqrt{1500}} = \frac{30}{38.7298} = 0.7746.$$

$r^2 = 0.60$ — the line explains 60% of the variance in y .

7 Multiple Linear Regression

Hyperplane and cost:

$$\hat{y} = w_0 + w_1 x_1 + \cdots + w_n x_n, \quad J(\mathbf{w}) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}_i - y_i)^2,$$

GD update $w_j \leftarrow w_j - \alpha \frac{\partial J}{\partial w_j}$.

7.1 Gradient of the MSE cost (derivation)

With $\hat{y}_i = \sum_k w_k x_{ik}$ (and $x_{i0} = 1$),

$$\frac{\partial J}{\partial w_j} = \frac{1}{2m} \sum_{i=1}^m 2(\hat{y}_i - y_i) \frac{\partial \hat{y}_i}{\partial w_j} = \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i) x_{ij}.$$

Hence the concrete update $w_j \leftarrow w_j - \frac{\alpha}{m} \sum_i (\hat{y}_i - y_i) x_{ij}$. ■

8 Lagrange Interpolation

8.1 Formula

$$f(x) = \sum_{j=0}^n y_j \prod_{\substack{k=0 \\ k \neq j}}^n \frac{x - x_k}{x_j - x_k}.$$

$(n+1)$ points $\Rightarrow$ at most an n -th order polynomial.

8.2 Proof

Assume an n -th degree polynomial

$$\begin{aligned} f(x) = & A_0(x - x_1)(x - x_2) \cdots (x - x_n) + A_1(x - x_0)(x - x_2) \cdots (x - x_n) \\ & + \cdots + A_n(x - x_0)(x - x_1) \cdots (x - x_{n-1}). \end{aligned} \quad (1)$$

Put $x = x_0$: every term except the first has a factor $(x - x_0)$ and vanishes,

$$f(x_0) = A_0(x_0 - x_1)(x_0 - x_2) \cdots (x_0 - x_n),$$

and $f(x_0) = y_0$ gives $A_0 = \frac{y_0}{(x_0 - x_1)(x_0 - x_2) \cdots (x_0 - x_n)}$. Putting $x = x_1, \dots, x_n$ similarly yields $A_1, \dots, A_n$. Substituting all A_k into (1) gives the stated $f(x)$. ■

8.3 Worked simulation — 3 points, estimate $f(2)$

Points $(x_0, y_0) = (1, 1)$, $(x_1, y_1) = (3, 9)$, $(x_2, y_2) = (4, 16)$ (sampled from $y = x^2$). Estimate $f(2)$.

$$L_0(2) = \frac{(2-3)(2-4)}{(1-3)(1-4)} = \frac{(-1)(-2)}{(-2)(-3)} = \frac{2}{6} = 0.3333,$$

$$L_1(2) = \frac{(2-1)(2-4)}{(3-1)(3-4)} = \frac{(1)(-2)}{(2)(-1)} = \frac{-2}{-2} = 1,$$

$$L_2(2) = \frac{(2-1)(2-3)}{(4-1)(4-3)} = \frac{(1)(-1)}{(3)(1)} = -0.3333.$$

$$f(2) = 1(0.3333) + 9(1) + 16(-0.3333) = 0.3333 + 9 - 5.3333 = 4.$$

Exact value $2^2 = 4$ — matches (the three points determine the exact quadratic). ■

9 Linear Interpolation

9.1 Derivation

Line through the two bracketing points $(x_2, y_2), (x_3, y_3)$: $y_2 = mx_2 + b$, $y_3 = mx_3 + b \Rightarrow m = \frac{y_3 - y_2}{x_3 - x_2}$. For the query point (x_m, y_m) on the same line, $y_m - y_2 = m(x_m - x_2) \Rightarrow m = \frac{y_m - y_2}{x_m - x_2}$. Equating the two slopes:

$$\frac{y_3 - y_2}{x_3 - x_2} = \frac{y_m - y_2}{x_m - x_2}. \blacksquare$$

Rule: always use the two tabulated points that *bracket* the query x .

9.2 Worked simulation

Table: $(x_2, y_2) = (10, 100)$, $(x_3, y_3) = (20, 400)$. Find y at $x_m = 13$.

$$\frac{400 - 100}{20 - 10} = \frac{y_m - 100}{13 - 10} \Rightarrow \frac{300}{10} = \frac{y_m - 100}{3} \Rightarrow 30 = \frac{y_m - 100}{3},$$

$$y_m - 100 = 90 \Rightarrow y_m = 190.$$

(Straight-line estimate between the two points.) ■

10 Newton's Divided-Difference Interpolation (v2)

10.1 Form and coefficients

$$P_n(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + \cdots + a_n \prod_{i=0}^{n-1} (x - x_i),$$

$$a_0 = y_0, \quad a_1 = \frac{y_1 - y_0}{x_1 - x_0} = f[x_1, x_0], \quad a_2 = f[x_2, x_1, x_0], \quad \dots$$

Recurrence (master formula):

$$f[x_0, \dots, x_{n+1}] = \frac{f[x_1, \dots, x_{n+1}] - f[x_0, \dots, x_n]}{x_{n+1} - x_0}.$$

10.2 Derivation of the first two coefficients

$P_0(x) = a_0$; at $x = x_0$: $P_0(x_0) = a_0 = y_0$, so $\boxed{a_0 = y_0}$.

$P_1(x) = a_0 + a_1(x - x_0)$; at $x = x_1$: $y_1 = y_0 + a_1(x_1 - x_0) \Rightarrow a_1 = \frac{y_1 - y_0}{x_1 - x_0} = f[x_1, x_0]$.

$P_2(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1)$; at $x = x_2$:

$$y_2 = y_0 + f[x_1, x_0](x_2 - x_0) + a_2(x_2 - x_0)(x_2 - x_1).$$

Rearranging, $\frac{y_2 - y_0}{x_2 - x_0} = f[x_1, x_0] + a_2(x_2 - x_1)$, hence

$$a_2 = \frac{f[x_2, x_0] - f[x_1, x_0]}{x_2 - x_1} = \frac{f[x_2, x_1] - f[x_0, x_1]}{x_2 - x_0} = f[x_2, x_1, x_0]. \blacksquare$$

10.3 Proof of the recurrence relation

Let $P_n(x)$ interpolate $x_0 \rightarrow x_n$ and $L_n(x)$ interpolate $x_1 \rightarrow x_{n+1}$ (both n -th order). Define

$$P_{n+1}(x) = \frac{(x - x_0)L_n(x) + (x_{n+1} - x)P_n(x)}{x_{n+1} - x_0}.$$

At $x = x_0$: the first term vanishes, $P_{n+1}(x_0) = \frac{(x_{n+1} - x_0)P_n(x_0)}{x_{n+1} - x_0} = P_n(x_0) = y_0$.

At $x = x_{n+1}$: the second term vanishes, $P_{n+1}(x_{n+1}) = L_n(x_{n+1}) = y_{n+1}$.

At an interior $x = x_k$: both $L_n(x_k) = P_n(x_k) = y_k$, so

$$P_{n+1}(x_k) = \frac{(x_k - x_0)y_k + (x_{n+1} - x_k)y_k}{x_{n+1} - x_0} = \frac{(x_{n+1} - x_0)y_k}{x_{n+1} - x_0} = y_k.$$

Thus P_{n+1} interpolates all $n+2$ points. Writing both P_n and L_n in Newton form and equating the coefficient of the top power x^{n+1} on each side gives

$$f[x_0, \dots, x_{n+1}] = \frac{f[x_1, \dots, x_{n+1}] - f[x_0, \dots, x_n]}{x_{n+1} - x_0}. \blacksquare$$

10.4 Worked simulation — build the table & polynomial

Points $(1, 1), (2, 8), (3, 27), (4, 64)$ (from $y = x^3$). All $\Delta x = 1$.

1st divided differences: $f[x_0, x_1] = \frac{8-1}{2-1} = 7$, $f[x_1, x_2] = \frac{27-8}{3-2} = 19$, $f[x_2, x_3] = \frac{64-27}{4-3} = 37$.

2nd: $f[x_0, x_1, x_2] = \frac{19-7}{3-1} = 6$, $f[x_1, x_2, x_3] = \frac{37-19}{4-2} = 9$.

3rd: $f[x_0, x_1, x_2, x_3] = \frac{9-6}{4-1} = 1$.

Top diagonal $\Rightarrow a_0 = 1, a_1 = 7, a_2 = 6, a_3 = 1$. So

$$P_3(x) = 1 + 7(x - 1) + 6(x - 1)(x - 2) + 1(x - 1)(x - 2)(x - 3).$$

Check at $x = 2.5$: $1 + 7(1.5) + 6(1.5)(0.5) + 1(1.5)(0.5)(-0.5) = 1 + 10.5 + 4.5 - 0.375 = 15.625 = 2.5^3$. **✓Reuse point:** adding $(5, 125)$ only appends one bottom row and one new coefficient a_4 ; $a_0..a_3$ are untouched. $\blacksquare$

11 Spline Interpolation (v2)

11.1 Equation-count bookkeeping

Linear	$P_i = m_i x + c_i$	$2n$ unknowns (each piece a line)
Quadratic	$P_i = a_i x^2 + b_i x + c_i$	$3n$ unknowns
Cubic	$P_i = a_i x^3 + b_i x^2 + c_i x + d_i$	$4n$ unknowns

$$\begin{aligned} \text{Quadratic: } 3n &= \underbrace{2n}_{\text{through pts}} + \underbrace{(n-1)}_{P' \text{ match}} + \underbrace{1}_{P''=0 \text{ at one end}}. \\ \text{Cubic: } 4n &= \underbrace{2n}_{\text{through pts}} + \underbrace{(n-1)}_{P' \text{ match}} + \underbrace{(n-1)}_{P'' \text{ match}} + \underbrace{2}_{\text{boundary}}. \end{aligned}$$

11.2 Worked simulation — quadratic spline, 3 points

Points $(0, 0), (1, 1), (2, 4)$. Two splines ($n = 2$): $P_1(x) = a_1x^2 + b_1x + c_1$ on $[0, 1]$, $P_2(x) = a_2x^2 + b_2x + c_2$ on $[1, 2]$. $3n = 6$ unknowns.

(1) **Through points ($2n=4$):** $P_1(0) = 0 \Rightarrow c_1 = 0$; $P_1(1) = 1 \Rightarrow a_1 + b_1 + c_1 = 1$; $P_2(1) = 1 \Rightarrow a_2 + b_2 + c_2 = 1$; $P_2(2) = 4 \Rightarrow 4a_2 + 2b_2 + c_2 = 4$.

(2) **Smoothness,** $P'_1(1) = P'_2(1)$ ($n - 1 = 1$): $2a_1 + b_1 = 2a_2 + b_2$.

(3) **Assumption** $P''_1(0) = 0$ (1): $2a_1 = 0 \Rightarrow a_1 = 0$.

Solve: $a_1 = 0 \Rightarrow b_1 = 1$ (from $a_1 + b_1 = 1$), $c_1 = 0$, so $P_1(x) = x$. Smoothness: $2(0) + 1 = 2a_2 + b_2 \Rightarrow 2a_2 + b_2 = 1$. From the two P_2 equations: subtract ($a_2 + b_2 + c_2 = 1$) from ($4a_2 + 2b_2 + c_2 = 4$) to get $3a_2 + b_2 = 3$. With $2a_2 + b_2 = 1$: $a_2 = 2$, $b_2 = -3$, $c_2 = 1 - a_2 - b_2 = 1 - 2 + 3 = 2$. So

$$P_1(x) = x \text{ on } [0, 1], \quad P_2(x) = 2x^2 - 3x + 2 \text{ on } [1, 2].$$

Check $P_2(2) = 8 - 6 + 2 = 4$. ✓

11.3 Cubic spline boundary conditions

Natural cubic: $P''_1(x_{\text{first}}) = 0$ and $P''_n(x_{\text{last}}) = 0$ (zero curvature at the ends). **Clamped cubic:** instead you are given the end slopes $\frac{dy}{dx}$ at the first and last nodes. Either choice supplies the final 2 equations so that $4n$ unknowns are fully determined.

Reproduce every boxed derivation from memory before the exam.